Data Preparation and Transformation in Power BI using Power Query Editor

1. Introduction

Data preparation and transformation are foundational stages in the data analysis process. In Power BI, these operations are performed primarily in the **Power Query Editor**, a powerful interface designed for cleaning, structuring, and shaping raw data before it is loaded into the Power BI data model.

The main objective of data preparation is to ensure that the dataset is accurate, consistent, and analysis-ready. It includes removing unnecessary data, correcting data types, handling missing values, renaming columns, and creating calculated fields. These tasks make the data suitable for visualization, reporting, and decision-making.

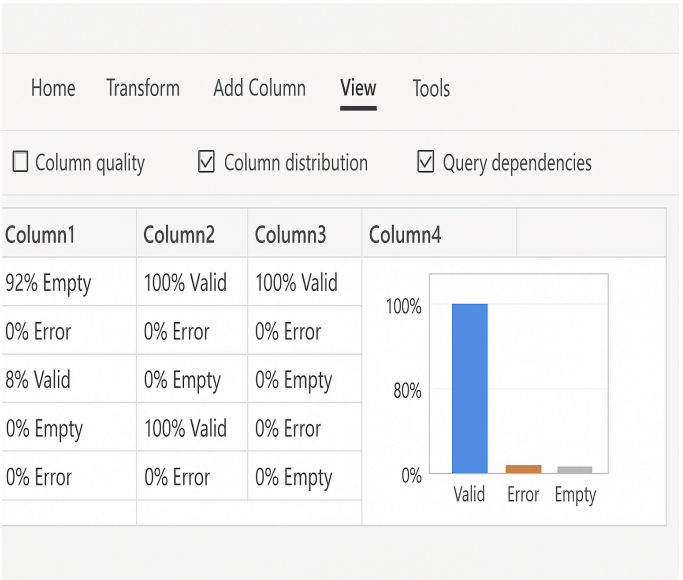
This note provides a detailed walkthrough of data cleaning and transformation steps using Power Query Editor, covering theoretical concepts, practical steps, and reasoning behind each operation.

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2. Understanding Data Quality and Column Distribution

Before performing any transformations, it is crucial to understand the **quality and distribution of data** in each column. Power BI provides two key options under the **View tab** in Power Query Editor:

- **Column Quality:** Displays the percentage of **Valid**, **Error**, and **Empty** values in each column.
- **Column Distribution:** Provides a quick visual representation of how data is distributed, helping to identify outliers, duplicates, and empty cells.

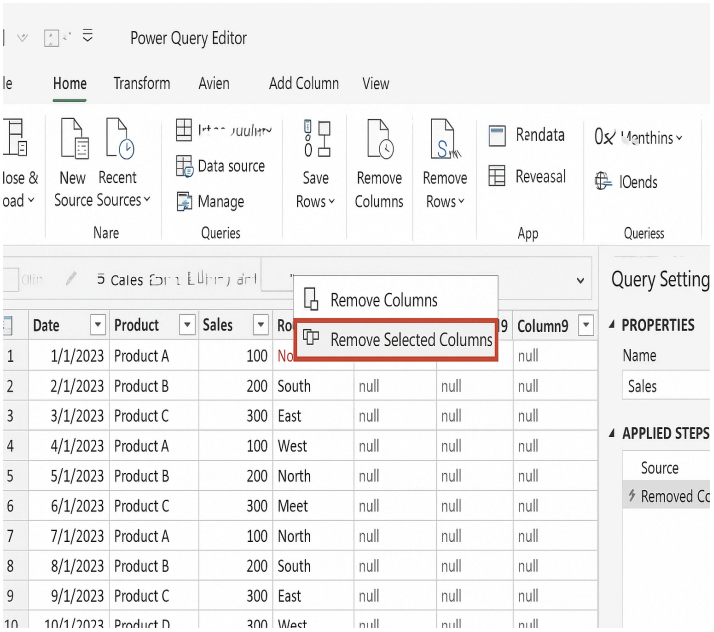


By enabling these options (using checkboxes in the View tab), users can quickly assess which columns contain missing or invalid data.

For instance, if the first column shows **92% empty values**, **0% errors**, and **8% valid entries**, it signals that this column requires attention before proceeding with analysis. Such insights guide decisions about whether to retain, clean, or remove specific columns.

3. Removing Unnecessary Columns

After assessing column quality, users often identify columns that are completely or largely empty. In the example provided, **columns 5, 8, and 11** are entirely null and therefore can be deleted.



Practical Steps to Remove Columns:

1. Select the columns to be deleted (using Ctrl + Click).
2. Go to the **Home tab** in Power Query Editor.
3. Click **Remove Columns → Remove Selected Columns**.

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This operation not only deletes the columns from the dataset but also adds a transformation step to the **Applied Steps pane**, ensuring that the change is recorded and automatically applied whenever new data is loaded.

4. Removing Unwanted Rows

In many datasets, the first few rows may contain metadata such as distributor names, summaries, or notes—data not relevant for analysis.

To remove the first **five rows** (or any number of top rows):

Steps:

1. Go to **Home tab → Remove Rows → Remove Top Rows**.
2. Enter the number of rows to remove (e.g., 5).
3. Click **OK**.

Optionally, you can create a **parameter** to dynamically control how many rows are removed. Parameters are useful for advanced scenarios where the number of rows may vary between datasets.

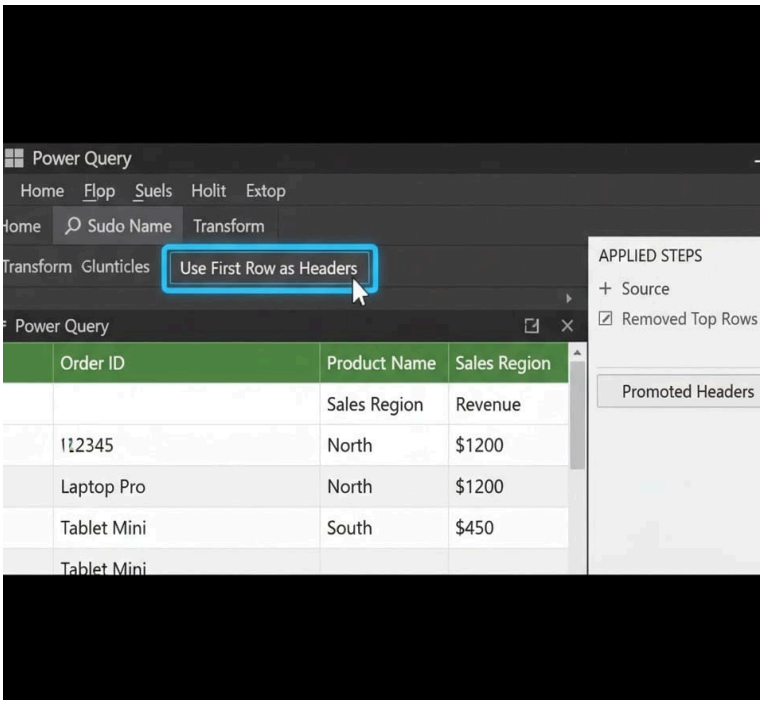
After this step, the dataset's first row often contains **column headers**, which must be properly recognized.

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5. Using the First Row as Headers

Once irrelevant rows are removed, the first row in the table usually represents **column names**. However, Power BI may still display them as "Column1", "Column2", etc. To fix this:

- Go to **Home tab → Use First Row as Headers**.



This transformation assigns meaningful names to each column. However, due to issues like merged cells in the source ERP data, column names may sometimes appear combined in a single column.

6. Correcting Column Header Issues

When column headers are merged or incorrectly assigned (e.g., multiple names combined in one cell), manual adjustment is necessary:

1. Double-click the column header to edit it.
2. Use **Ctrl + X** to cut unwanted portions of the header.
3. Paste them into their respective correct columns.
4. Clean up extra spaces, hyphens, or unwanted characters.
5. Press **Enter** to apply the changes.

This process should be repeated for all affected columns until each has a distinct and accurate name. Clean and meaningful column headers make data analysis more intuitive and minimize errors in later stages.

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7. Verifying and Assigning Data Types

Each column in Power BI has an associated **data type**, indicated by an icon next to the column name (e.g., ABC for text, 1.2 for decimals, 123 for whole numbers). Power BI typically detects data types automatically, but they should always be verified:

Common Data Types in Power BI:

- **Text (ABC)** – for categorical data such as names, categories, or labels.
- **Whole Number (123)** – for integer values.
- **Decimal Number (1.2)** – for fractional or monetary data.
- **Date/Time** – for temporal data fields.
- **True/False** – for binary or logical fields.

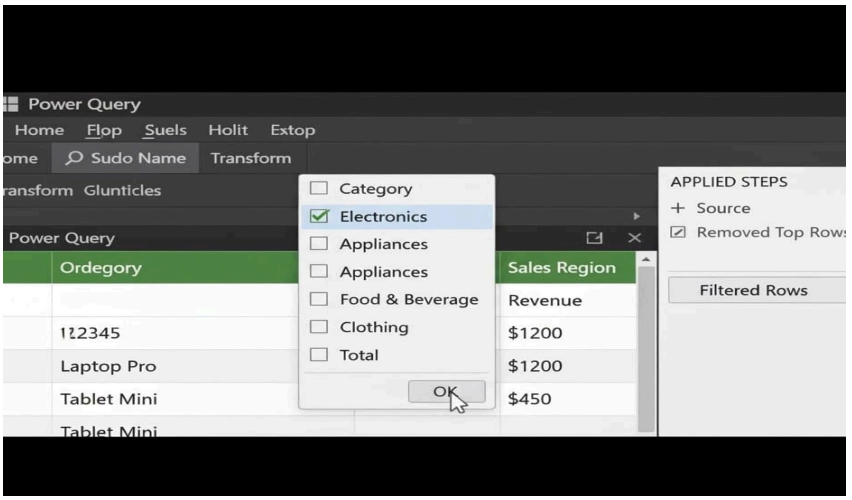
If Power BI assigns an incorrect data type, it can be changed manually:

- Click the data type icon → choose the correct type (e.g., Text, Whole Number, or Decimal).

Accurate data typing ensures that mathematical operations, sorting, and aggregations work as intended.

8. Filtering Out Unnecessary Values

In some columns, such as **Pack Type** or **Category**, certain rows may contain non-relevant labels like "Total" within the data. These should be filtered out to prevent incorrect aggregations.



Steps:

1. Select the column.
2. Click the **Filter Dropdown**.
3. Uncheck the unwanted value (e.g., "Total").
4. Click **OK**.

This action removes unnecessary rows and retains only valid data points.

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9. Handling Null Values with Fill Down Operation

In real-world datasets, columns often contain **null values** where the same value should be repeated for multiple rows (especially in hierarchical or grouped data). For example:

- The *Outlet* name or *Category* might appear once and the following rows are null until the next outlet appears.

To fill these nulls with the value from the previous row:

Steps:

1. Select the target column (e.g., Outlet).
2. Go to **Transform tab → Fill → Fill Down**.

This replaces nulls with the value directly above them.
Repeat the same process for other columns such as Category.

Conceptual Understanding:

- **Fill Down** fills missing values downward using the preceding value.
- **Fill Up** does the reverse—fills missing values upward using the next available value.

After performing these operations, the dataset becomes more structured and consistent, making it suitable for visualization and analysis.

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10. Understanding Transform Tab vs Add Column Tab

Both **Transform** and **Add Column** tabs in Power Query Editor offer similar options (e.g., text formatting, calculations, or conditional logic). However, they differ fundamentally in their purpose:

Feature	Transform tab	Add column tab
Purpose	Modifies data within the same column	Creates a new column for the result
Effect	Overwrites existing column values	Retains original column and adds new one
Example	Changing text to uppercase in the same column	Creating a new uppercase column

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Example:
When we applied **Fill Down** earlier using the Transform tab, it modified the existing column without adding a new one. In contrast, creating a new calculated column uses the Add Column tab.

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11. Creating a Calculated Column (Add Column Example)

Let's consider creating a new column, **Revenue per Quantity**, by dividing Gross Revenue by Sales Quantity.

Steps:

1. Select the **Gross Revenue** column.
2. Hold **Ctrl** and select **Sales Quantity**.
3. Go to **Add Column → Standard → Divide**.

This operation generates a **new column** where each row equals:

Revenue per Quantity=Gross RevenueSales Quantity\text{Revenue per Quantity} = \frac{\text{Gross Revenue}}{\text{Sales Quantity}}Revenue per Quantity=Sales QuantityGross Revenue

For example:

- Row 1: 1205.26 ÷ 6.13 = 196.61
- Row 2: 6133.3 ÷ 10 = 613.33

The new column automatically appears in the dataset and in the Applied Steps pane.

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12. Handling Errors and Formatting the New Column

Sometimes, the denominator (Sales Quantity) may contain **zero values**, leading to division errors. These can be resolved by filtering out zero values:

Steps:

1. Select the **Sales Quantity** column.
2. Click the filter dropdown.
3. Uncheck **0** and **null** values.
4. Click **OK**.

Now, you can format the new column:

- Change **Data Type** → *Fixed Decimal Number*.
- Rename column → *Revenue per Quantity*.
- Apply **Currency format** if appropriate.

This ensures uniform decimal places and error-free numeric values.

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13. Text Formatting Operations

Power BI allows text transformations such as changing case, trimming spaces, or removing prefixes/suffixes. These are accessible under both **Transform** and **Add Column** tabs.

Example: Capitalizing Each Word

1. Select the **Outlet** column.
2. Go to **Add Column → Format → Capitalize Each Word**.

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A new column is created with formatted text (e.g., “8 BS CAFÉ” becomes “8 Bs Café”).

If instead you use **Transform → Format → Capitalize Each Word**, the transformation happens **in-place** without creating a new column.

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14. Managing and Reordering Columns

After transformations, you may wish to rearrange columns for better readability. Power Query allows column reordering through:

- **Drag and Drop** in the data preview pane.
- Right-click column header → **Move → To Beginning / To End**.

You can also remove intermediate transformation steps (like reordering) from the **Applied Steps** panel to clean up the query history.

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15. Applied Steps Pane – Understanding Query History

Each operation performed in Power Query (like removing rows, renaming columns, or adding calculated fields) is automatically recorded as a **step** in the *Applied Steps* pane.
This provides:

- A transparent record of transformations.
- The ability to revisit, modify, or delete specific steps.
- Automation—Power Query re-applies all steps when the data source is refreshed.

This stepwise approach forms the **query logic**, expressed in Power BI's **M Language** (Mashup Language).

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16. Overview of M Language

Every transformation in Power Query is internally written in **M Language**, a functional scripting language.
For instance, when calculating *Revenue per Quantity*, the M formula might look like:

```
= Table.AddColumn(#"Previous Step", "Revenue per Quantity",
each [Gross Revenue] / [Sales Quantity], type number)
```

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This expression:

- Creates a new column named "Revenue per Quantity".
- Performs division for each row.
- Assigns the result as a numeric value.

M language scripts are auto-generated but can also be edited manually for advanced data manipulation.

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17. Summary of Key Transformation Techniques

Operation	Tab Used	Purpose	Practical Use
Remove Columns	Home	Delete irrelevant columns	Clean dataset
Remove Rows	Home	Delete top or specific rows	Exclude unwanted data
Use First Row as Headers	Home	Promote first row to headers	Structure dataset
Change Data Type	Transform	Assign proper types	Ensure accurate analysis
Filter Values	Column Menu	Exclude unwanted data (e.g., “Total”)	Refine dataset
Fill Down	Transform	Replace nulls with previous value	Maintain continuity
Add Column	Add Column	Create new calculated field	Derive insights
Format Text	Transform/Add Column	Clean textual data	Improve readability

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18. Best Practices in Data Preparation

1. **Inspect data quality first** using the View tab.
2. **Rename columns meaningfully** for easier reference in visuals.
3. **Assign correct data types** to prevent calculation errors.
4. **Remove or filter out null and irrelevant data** cautiously.
5. **Use Fill Down or Fill Up** to handle hierarchical data.
6. **Leverage Add Column** for derived measures instead of overwriting.
7. **Regularly check Applied Steps** to maintain query integrity.
8. **Test transformations** with small data samples before full application.

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19. Conclusion

Data preparation in Power BI is not just a mechanical process—it's an analytical exercise that determines the accuracy and reliability of insights derived later.
By systematically cleaning, transforming, and structuring the data in Power Query Editor, users can ensure their datasets are accurate, consistent, and meaningful.

The **View tab** provides an overview of data health, the **Home and Transform tabs** handle structural and in-place modifications, while the **Add Column tab** enables creation of new analytical measures.
Together, these tools enable analysts to move confidently from raw, unstructured data to powerful, insight-ready datasets—forming the foundation for effective visualization and reporting in Power BI.

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